INSTITUTE OF SONOLOGY

Royal Conservatoire The Hague

IanniX

Composing with Algorithms
http://www.bjarni-gunnarsson.net

"Just as there is no measurement of sound, or more accurately, no generally agreed upon such measure, there is no agreement and consequently tremendous variation in the graphical description of sound. From concréte-like descriptive morphologies to notational explorations, to technical attempts, to synaesthetic experiments [..] in fact to every conceivable use and misuse of graphics to visualize sound, there is almost complete non-concordance on what we see when we hear, or what we should see when we hear."

(Larry Polansky)

UPIC

UPIC

A computer music system for creating sounds using a graphic interface. The first UPIC machine was made by the ***CEMAMu*** (Centre d'Etudes de Mathématique et Automatique Musicales) in 1977.

An important update came in 1991, making the system fully ***interactive***. UPIC was finally released as Windows software in 2001. This release required none of the external hardware.

Among the composers having used ***UPIC*** are Julio Estrada, Gerard Pape, Bernard Parmegiani, Jean-Claude Risset, Daniel Teruggi and Iannis Xenakis.

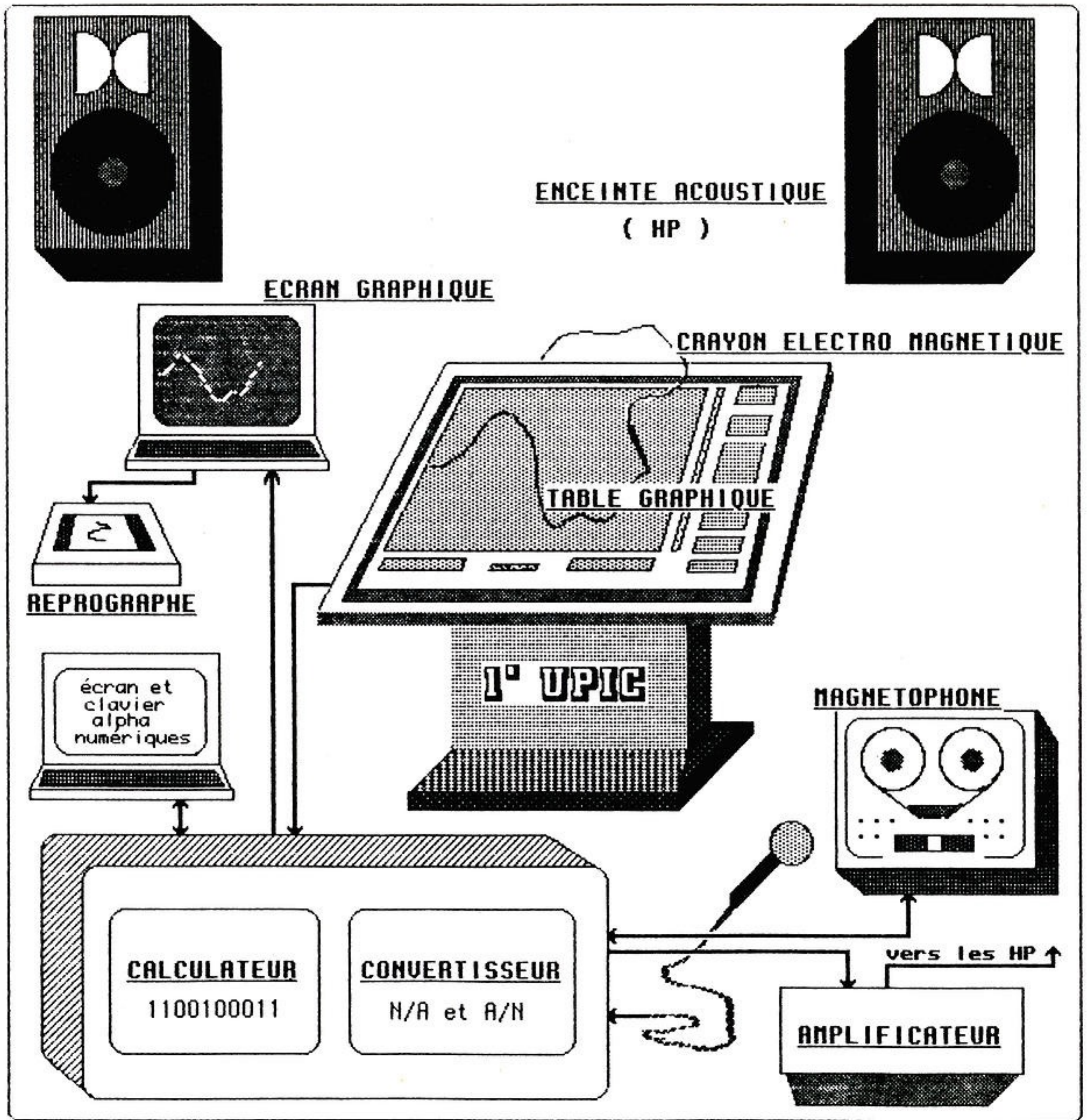
UPIC

The compositional elements are designed with an ***pen*** on a large electromagnetic drawing ***board***. A mouse could also be used.

These elements exist as ***lines*** (or other graphic shapes) and controls include ***waveforms***, ***envelopes*** and "***arcs***" (notes). The shapes are combined on a ***page*** that is then read to produce sound.

The shapes can also be used for ***modulation*** (such as the speed a page is read at).

UPIC



ENCEINTE ACOUSTIQUE
(HP)
ECRAN GRAPHIQUE
CRAYON ELECTRO MAGNETIQUE
TABLE GRAPHIQUE
REPROGRAPHE
1° UPIC
écran et clavier alpha numériques
MAGNETOPHONE
CALCULATEUR
1100100011
CONVERTISSEUR
N/A et A/N
vers les HP
AMPLIFICATEUR

UPIC

The UPIC shapes are drawn via a tablet which empathises the ***gestural impact*** the software wants to achieve. One can draw ***curves*** for several things with the UPIC but the ***pitch parameter*** is the one that really makes it special and gives it an unstable and rough quality.

Creating original or strange ***glissandi*** sounds is fairly straightforward using the system.

The UPIC system can be seen as a sequence of events of ***spatial exploration***. A UPIC page changes in space in a certain order. Sequence can however also be mapped to space instead of time.

UPIC

Gesture : *"A movement of a part of the body to express an idea or meaning. An action performed to convey one´s feelings or intention."*

Sound as a connection to ***physical movement***.

Drawing as a method for creating interesting musical forms.

Upic contains at the same time a precise ***representation*** of a sound as well as being the ***score*** of a piece.

UPIC can be seen as a sequencer for events of ***spatial exploration***.

UPIC

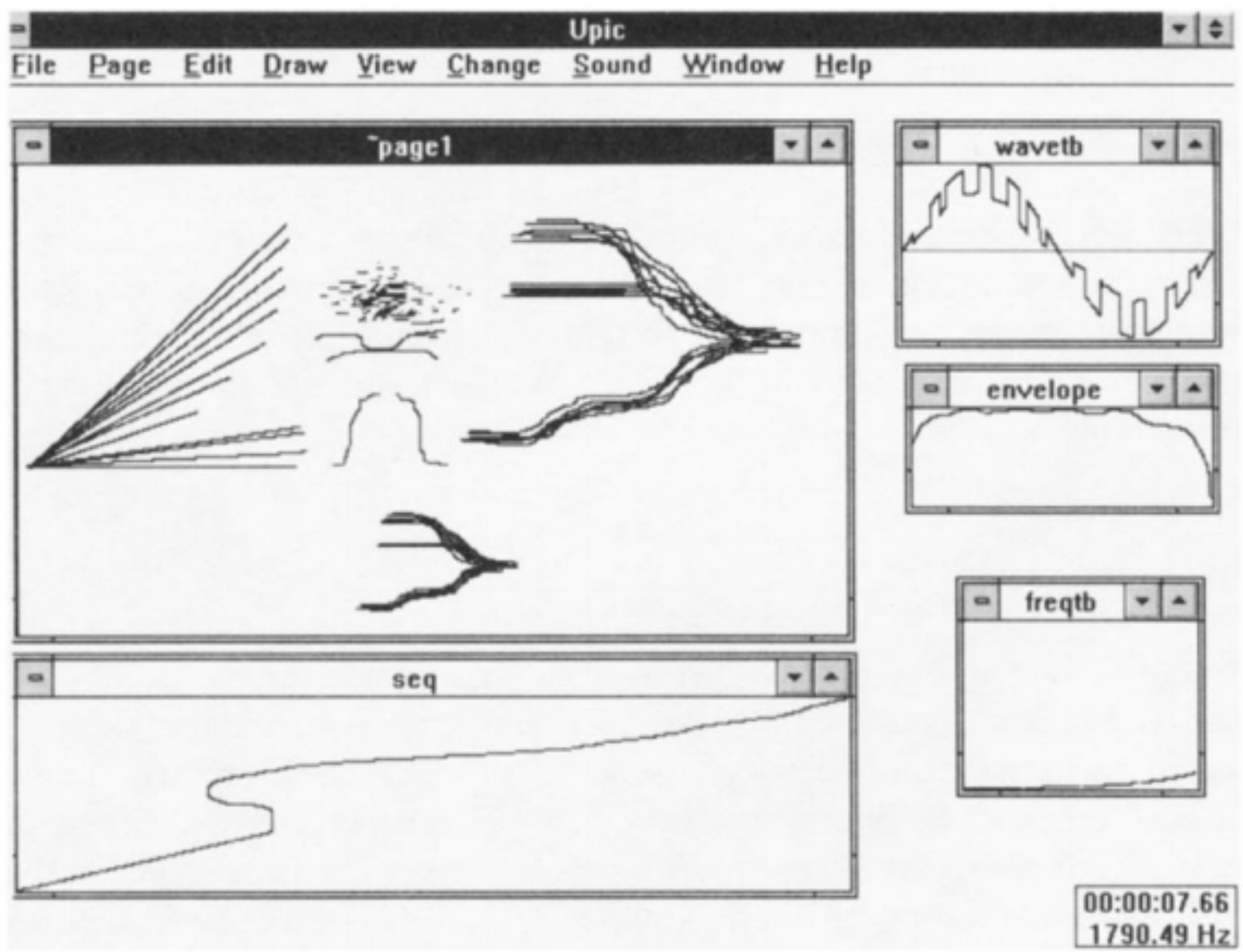
The exchange between ***thought*** and ***hearing*** was considered essential and real-time operation a first design requirement.

Another important idea was that a composer could work on the ***same time*** at many ***different levels*** at once. Furthermore all aspects related to a composition should be contained within the system (symbols or syntax).

Mapping was also represented using the arcs. For example one arc could play a sine glissandi while another line would define the pitch space the first one maps to (also done with a frequency table).

UPIC

'The Sequence Object' could be used to program jumps within reading points of a page. Here the reading cursor jumps to different parts at a specific interval.



UPIC

UPIC - Education

Xenakis wanted everybody to be able to "*compose*" with the ***UPIC***. He introduced it to dancers, kids, computer-oriented people and musicians. For example, during the decade of 80's and 90's many children experienced the UPIC environment in Greece and France.

UPIC

“Another idea was to let the composer control and create all aspects of the composition: sound, symbols, syntax, and so forth. This means that the system should not impose predefined sounds, predefined compositional process, predefined structures, and so on. It is essential for the creative mind that ideas not go through theories or limitations that might not suit the composer.”

“Children may draw a fish or a house and listen to what they have made and correct it. They can learn, progressively through designing, to think musical composition without being tormented by solfège or by incomplete mastery of a musical instrument.... But as they are led to construct rhythms, scales, and more complex things, they are also forced to combine arithmetic and geometric forms: music. From whence comes an interdisciplinary pedagogy through playing."
(Iannis Xenakis)

Mycenae Alpha

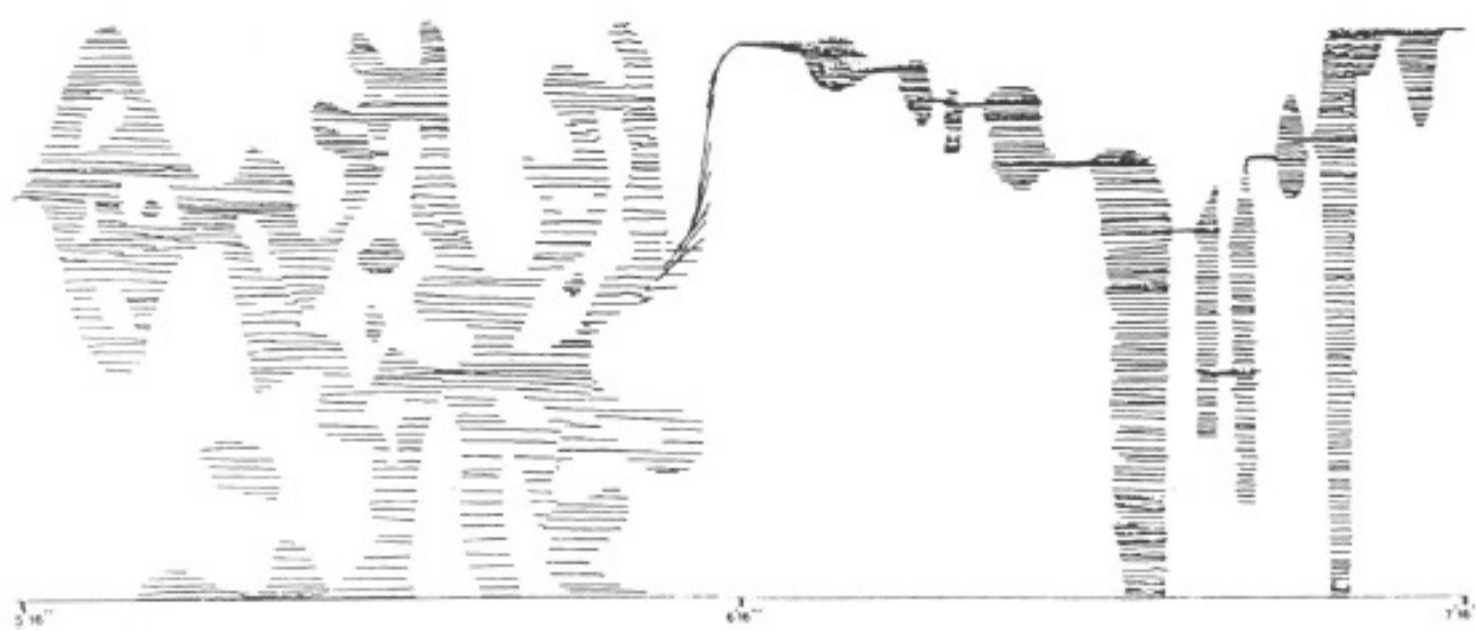
Interesting sounds can be created by designing noisy ***waveforms***, complex ***envelopes*** and layering ***lines***.

Xenakis was used to sketching his music on graph paper and the UPIC approach came naturally to him. The first piece created with the system was ***Mycenae Alpha*** completed in the summer of 1978 and composed for the Polytope de Mycènes, an outdoor event in Mycenae in Greece.

The music consists of noisy and dense ***clusters*** and the UPIC ***pages*** are designed to be visually compelling. A basic pattern is to move from ***complex textures*** to more ***constant clusters*** and back again to a complex sound.

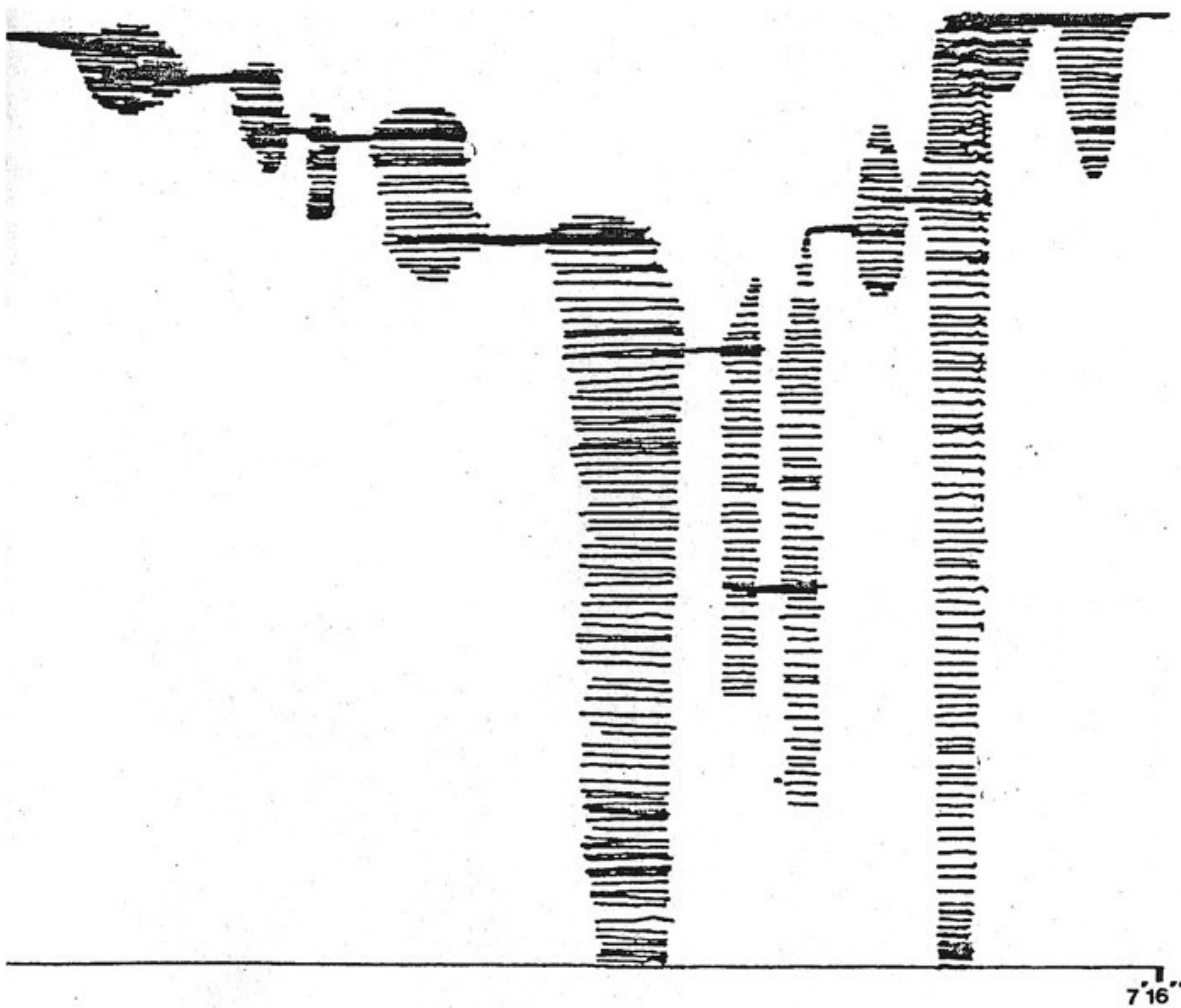
Mycenae Alpha

The initial version of UPIC did not allow overlapping of pages and Mycenae Alpha is simply a succession of different screens. Xenakis created ***twelve graphic/sonic entities***. Two of these are being repeated so the in total there are fourteen pages (screens).

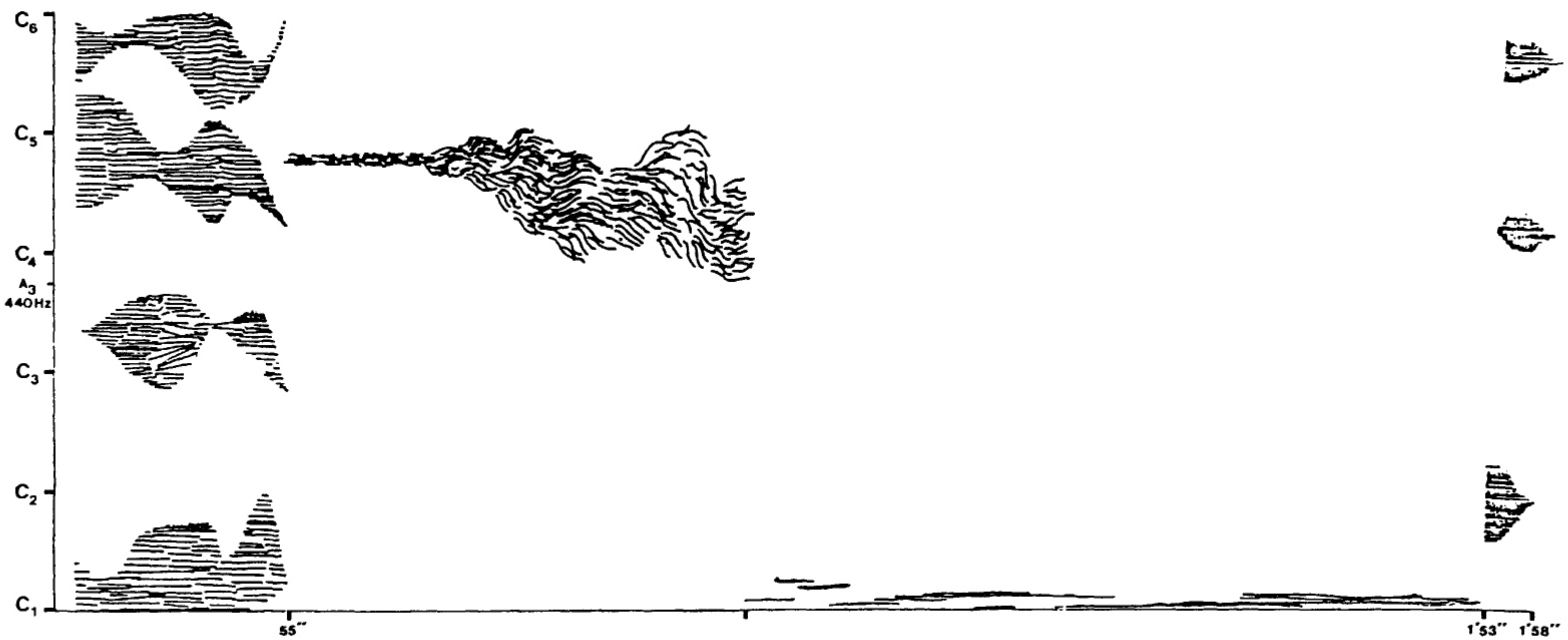


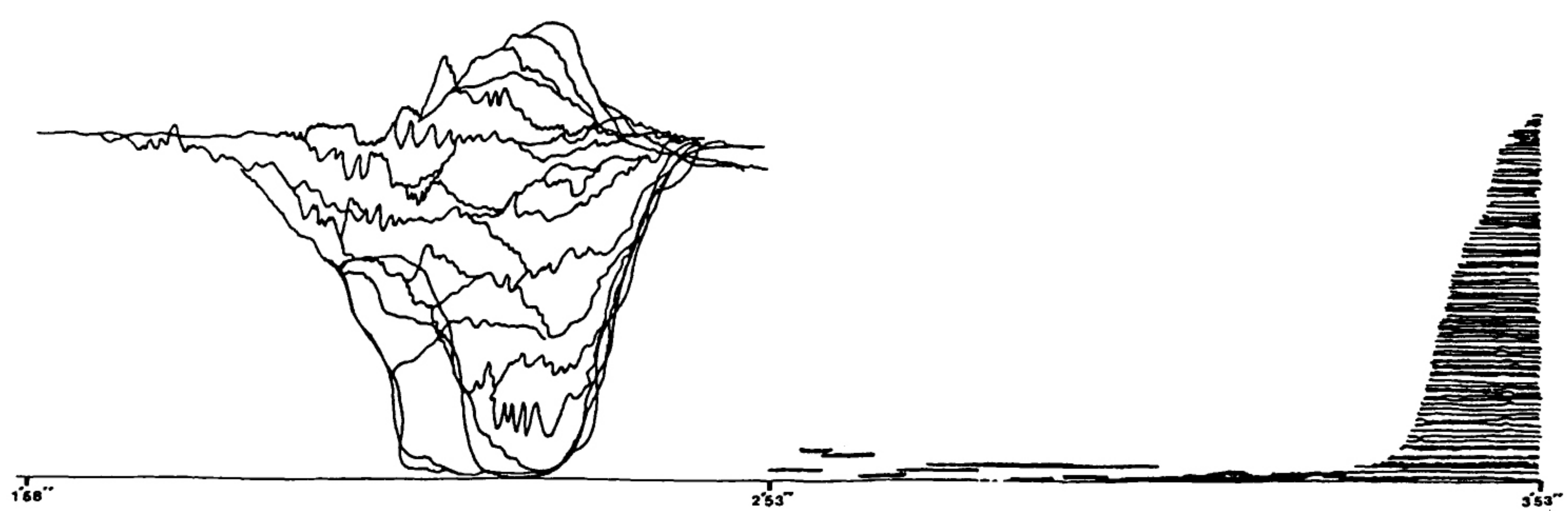
Mycenae Alpha

Mycènes alpha (1978), 6'16"-7'16"
Source : "partition", éditions Salabert



Mycenae Alpha





Mycenae Alpha

"A structural distinction can be made between complex sonorities created by means of masses of relatively stable note segments, and other sonorities created from dynamic arcs. Obviously, with no instrumental or procedural limitations, it is very easy to design complex glissandi on the UPIC merely by picking up the electromagnetic pen and tracing them onto the design board. What is not shown in the graphic score are the waveforms used for the individual timbres nor the dynamic envelopes for each note. While Mycenae alpha is remarkable for demonstrating the innovations of the UPIC system, there is no denying that, sonically, it is less interesting than La Légende d'Eer. One reason is that the waveforms used to create timbres on the UPIC are static. No matter how complex the waveform is, the computer will simply read through its limited digital representation and then repeat this process at a rate corresponding to the Hertz value (cycles per second) of the note to which the waveform is attached."

(James Harley, Xenakis: His Life in Music)

Xenakis - Mycenae Alpha (1978)

Haswell and Hecker

In 2004 ***Russell Haswell*** and ***Florian Hecker*** recorded material at CCMIX using the UPIC. They later released two albums and made several multi channel 'UPIC Diffusion Sessions' based on this material.

Many of the events included ***laser*** and ***stroboscopic*** lighting.

Haswell & Hecker - Movement 4

Blackest ever Black (2007)

Haswell and Hecker

"Haswell and Hecker's engagement with the project came from their belief that its potential has barely been tapped. Rather than concentrate on simple forms of mark-making and the sounds they produce, they experimented with different types of visual material – tracing onto the UPIC tablet images ranging from news photographs of disasters and atrocities, through depictions of the natural world to microscopic images of molecular structures (such as that which makes up 'the blackest ever black', a coating for telescopes that is purportedly the least reflective material on Earth). Despite the limitations of the system (it won't, for example, allow a full circle to be drawn – a sound that occurs over time can't, after all, travel back on itself), their research was in the spirit of stochastic exploration Xenakis advocated. Haswell and Hecker took the material they recorded at CCMIX and developed it for a diffusion system. No mere PA or surround sound set-up, a diffusion system is a multiple set of speakers that distribute sound through space, and allow for a high degree of real-time control over volume levels, equalization and, most importantly, spatial placement by the operator or performer at the mixing desk."

(Frieze Magazine)

BG - Pice (2006)

Iannix

Iannix

'IanniX is a graphical open source sequencer, based on Iannis Xenakis works, for digital art. IanniX syncs via Open Sound Control (OSC) events and curves to your real-time environment.'

Iannix

Official website:

http://www.iannix.org

GitHub:

https://github.com/iannix/IanniX

Installing Iannix:

http://www.iannix.org/en/download-iannix/

Configuring with SuperCollider (very basic video)

https://www.youtube.com/watch?v=TbBvobXA7ao

Iannix

Iannix has three main kinds of objects:

Triggers (events)
Curves (spatial trajectories)
Cursors (playheads)

Triggers (discrete) and *Cursors* (continuous) send ***messages*** that can be of various kind (OSC, MIDI, TCP). The messages can be configured to send different kinds of information (*id, location, cursor of a trigger* and many more).

Messages can also be ***internal*** and used to start other objects, change the global speed, camera etc.

Iannix

Important shortcuts when using Iannix:

- ***Click and move*** : Navigate the score.
- ***Double click object*** : Edit the object message(s).
- ***Mouse wheel*** : Movement for zooming.
- ***Alt click + mouse move*** : Change the camera perspective.
- ***Alt mouse wheel*** : Change the camera distance.
- ***Alt double click*** : Reset the camera.
- ***Shift double click:*** Forces an object to send its messages.
- ***CMD alt*** : Forces snap-to-grid for objects.

Iannix

In order to use the attached files, copy all the ***.iannix** files contained with this class to the ***Iannix documents folder*** (Documents/Iannix on OSX).

After rebooting Iannix these should appear under the ***files tab*** in the selection screen on the right side of the program.

The synths used for the examples are contained in the file ***PMB10 - Iannix.scd***

The examples use port ***57120*** (NetAddr.langPort = 57120). Both Iannix and SuperCollider need to be set using the same port.

Iannix - Scripting

A Iannix script consists of six main ***functions*** and more can be included.

Three of these functions are created ***automatically*** by Iannix once something is created/edited/deleted using the GUI or from external programs. The other three functions can be ***overwritten*** for customized behaviour.

The programming language is ***Javascript***. It shares some attributes with SuperCollider such as first order functions and a dynamic attitude.

Iannix - Scripting

```
// The good section for asking user for script global variables
function askUserForParameters()

// Stores customized user code. Is never overwritten
function makeWithScript()

// Is called when an incoming message is received
function onIncomingMessage()

// Stores all the operations made through the graphical user interface.
function madeThroughGUI()

// operations made by other softwares through one of the IanniX interfaces.
function madeThroughInterfaces()

// is called last and allows to modify a hand-drawn score
function alterateWithScript()
```

Iannix - Examples 1&2

PMB.Example1. A line with cursor that passes a trigger for generating a synth combined with a circle that sends continuous data to a filtered noise.

PMB.Example2. Two lines, each with a cursor.The first one contains two triggers as well that each play a sound that is modified by the cursors.The cursor on the right loops, the cursor on the left starts the one on the right both contain custom curves.

Iannix - Examples 3&4

PMB.Example3. A curve generated by a formula coupled with a simple line that controls the overall tempo.The SuperCollider system also sends OSC over to start the Iannix patch.

PMB.Example4. Generated lines, cursors and triggers.The amount is determined by the user input that runs when the patch is started.A custom function is defined that generates the lines and cursors.

Exercises

```
(
    NF(\iop, {|freq=78, mul=1.0, add=0.0|
        var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
        var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
        var out = DFM1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
        HPF.ar(out, 40)
    }).play;
)

(
    NF(\dsc, {|freq = 1080|
        HPF.ar(
            BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
            SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
            LFNoise1.ar([12,14,10]).range(100,900),
            SinOsc.ar(20).range(9,11)
        ), 80)
        ;
    }).play;
)
```

```
var <>pindex, <>cindex;

initialize {
    if(pindex.isNil, { pindex = 1000 });
    if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
    pindex = 1000;
    (this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
    if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
    if(index.isNil && pindex.isNil, {
        this.initialize();
    });

    pindex = pindex + 1;
    this[pindex] = \filter -> process;
}

control {|process, index|
    var i = index;

    if(i.isNil, {
        this.initialize();
        cindex = cindex + 1;
        i = cindex;
    });

    this[i] = \pset -> process;
}
```

```
(
    NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
        var trig, seq, freq;
        trig = Dust.kr(rate);
        seq = Dwhite(freqMin, freqMax, inf).midicps;
        freq = Demand.kr(trig, 0, seq);
        HPF.ar(FMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
        LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
    }).play;
)
```

Exercises - UPIC

1. Create a number of different lines and curves where each starts, modifies and finally stops a synthesis process in SuperCollider.

2. Compose a separate trajectory for each parameter of a synthesis process in SuperCollider (amplitude, frequency etc.) At the and end of each trajectory a new path for the parameter should be triggered with finally the option to return to the initial one once completed (circular repetition).

3. Generate a collection of shapes with Javascript where all tend to gravitate to a common attractor or area on the page.

4. Generate a collection of lines where each contains multiple triggers and a cursor for playback.Additionally a separate shape should be created that controls the overall speed of Iannix.

Exercises - Iannix

1: Glissando and Envelope. Draw a diagonal line in the FREQUENCY panel , an ascending glissando. In AMPLITUDE, draw a triangle: silence at the edges, loud in the middle. Play it. Then try the opposite direction.

2: Timbre as Shape. Keep frequency and amplitude flat. Focus only on the WAVEFORM panel. Try each preset, then draw your own shapes: zigzags, bumps, spikes. Each shape i s the timbre. Can you draw something that sounds noisy?

3: Sound Drawing. Using all three panels, try to draw recognisable sounds. With just three curves, you can suggest concrete sounds. Xenakis called t he direct correspondence between drawing and hearing "sound design by hand."

4: Sound Mass (Multi Voice). Draw several lines clustered in a narrow frequency band : a dense cloud. Clear and try the opposite: lines starting from one point, fanning outward. Then try convergence: scattered lines meeting at a single point.

5: Metastaseis (Multi Voice). Xenakis's Metastaseis (1954) used string glissandi to build continuous sound surfaces. Draw lines fanning upward from a shared starting pitch, then others gliding downward to converge. Change the waveform to something richer. Listen to where lines cross and where they spread apart.